# Young-Sang Lee

### Aspiring game server programmer, M.S. student

github.com/YSL-RyuDo | ysl-ryudo.github.io

I am interested in multiplayer server architecture and state synchronization.

I am pursuing an M.S. in Games (Engineering) at Hongik University, Sejong Campus.

## Applied technical skills

C++ / Windows networking, IOCP, Winsock, sessions, packet dispatch, resource pools and DB worker separation (VAIS, Arcana Duel, Slap Squad).

C# / Unity: character control, weapons and spells, match UI, branching dialogue, puzzle rules and content editors.

Python / experiments, C2PA validation, perceptual-hash retrieval and simulated consumption analysis; test-source and recorded-result review.

## Selected project experience

### CodeName VAIS

Mar 2026 ~ Present, 10 members: 5 software, 5 graphics

C++ server, Unity game flow, UI and tooling | C++, Windows IOCP, C#, Unity

Implemented IOCP connection handling, session management and packet dispatch; connected room/match lifecycle, career/profile, shop and social services to Unity client contracts.

Separated reconnect presentation recovery from gameplay execution: ReconnectJournal keeps persistent state and excludes replay of damage, rewards and player commands, so restoring the view does not repeat those actions.

Built Unity selection/loading presentation, profile stat and medal editing, friend and reward screens, and scene-data exporters. Integrated Sentinel abilities with network state.

Case study | Server / Client, Public repository

### Arcana Duel

Aug 2025 ~ Feb 2026, 2 members: 1 programmer, 1 graphics

Programming: Unity gameplay and C++ server | C++, IOCP, DB worker, Unity

Built the C++ IOCP server with receive/send workers and reusable session, buffer and overlapped-I/O resources; implemented the Unity game/network integration.

Assigned database requests to a separate DB worker, separating database execution from network completion handling.

Implemented four-slot deck editing, movement/status effects, projectile and field spells, cooldown HUD and victory/defeat flow. Separated local card presentation from runtime combat values and refreshed decks after accepted changes.

Case study | Server / Client, Public repository

### Slap Squad

Mar 2025 ~ Feb 2026, 8 members: 2 programmers, 6 graphics

Gameplay, C++ server and client integration | C++, Winsock, C#, Unity

Implemented the C++ Winsock server's account, room, shop and profile flows with connection threads and CSV-backed data; integrated the Unity client with the protocol.

Implemented seven weapon behaviors and worked on movement, balloon/map interaction, match results, rewards and emotes. Changed ShopUI ownership confirmation to follow returned item data.

Developed a CNN study of fixed versus demand-adjusted pricing from the game's item-pricing question; presented it at I2P-CON, then refined the research for the CISC-W 2025 poster.

Case study | Client / Server, Public repository

## Education & research experience

### 2026.09 ~ Present | M.S. student, Department of Games (Engineering)

Hongik University, Sejong Campus

### 2021 ~ 2026.08 | Bachelor's degree, Game Software

Hongik University, Sejong Campus

### 2025.03 ~ 2026.08 | Undergraduate researcher, Game Software

Hongik University, Sejong Campus

## Research & presentations

### CISC-W 2025, First author, poster presenter

2025 Winter Conference of the Korea Institute of Information Security and Cryptology

27 ~ 28 Nov 2025, Konjiam Resort, Gwangju, Korea

Consumption Pattern Analysis of In-Game Pricing Policies Using Convolutional Neural Networks

Compared pricing policies using 400 simulated players over 30 days, repeated 20 times. CNN consumption-pattern classification accuracy was 96%.

### WISA 2026, First author, poster presenter

27th World Conference on Information Security Applications

26 ~ 28 Aug 2026, MAISON GLAD, Jeju, Korea

A Verification Method for Game Image Assets Using C2PA Manifest Validation and Perceptual Hash-Based Similarity Analysis

Evaluated C2PA validation and four perceptual hashes across 1,150 validation cases and 4,400 retrieval rows. Published code, results, paper and poster.

### ICEA 2026, Accepted, presentation planned in Kaohsiung, research extensions in progress

International Conference on Intelligent Computing and its Emerging Applications

19 ~ 22 Nov 2026, Kaohsiung, Taiwan

Server-Side Multi-Parent Provenance Reconstruction in Heterogeneous Game MOD Packages

Research on multi-parent provenance retrieval using code, image and structured-resource features. Extending the accepted study while preserving its evaluation baseline.

## Leadership & knowledge sharing

### 2025 inaugural I2P-CON, Operations Team Lead & Speaker

Recruited the organizing team and led operations for the inaugural student-led game-development conference, jointly hosted by I2P and O2CUBE.

Presented the CNN pricing-policy research in “Designing Game Economies with Data.” Subsequently refined the research for the CISC-W 2025 poster presentation.

Produced a conference publication and distributed it to current students and alumni. The official playlist preserves eight talks.

My talk | I2P-CON

### O2CUBE, Software Team Lead, 2025

Served as Software Team Lead in 2025. The C programming and computer architecture study sessions ran from March to June. Taught C fundamentals through arrays and led the semester's computer architecture study.

## Additional engineering

DeathGuider: facilities and party preparation. Cylinder: branching dialogue, inventory and puzzles. DumbGuys: movement and obstacle gameplay. RE:BOUND and Moonlit Hotel: game progression and content editors. PitchLab: pitch targeting, audio buffering and recording.

I participate in software development at Hongik University's Game Content Copyright Technology Global Lab (GCT Lab).
GCT Lab